This is an excellent foundation for a dissertation project, combining advanced deep learning models for classification with a robust MLOps framework. Below is a structured outline and proposed content for your M.Tech dissertation thesis at VNIT Nagpur, drawing directly from the provided code and dataset sources.

Proposed Thesis Title

MLOps-Driven Comparative Analysis of Hybrid Deep Learning Architectures for Passive Human Identification using Wi-Fi Channel State Information (CSI)

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1. Introduction and Problem Statement (Chapter 1)

1.1 Background and Motivation

The demand for **passive, non-intrusive human identification (ID)** systems, particularly in high-security zones such as ISRO, NASA, and Parliament House, is growing. Traditional biometrics (like faces or fingerprints) often require active participation and are susceptible to spoofing.

This project focuses on leveraging **Wi-Fi Channel State Information (CSI)** as a sensing mechanism. CSI captures fine-grained channel variations that are highly sensitive to human presence and, critically, to individual movement signatures. By classifying these signatures, the system aims to identify a specific subject passively.

1.2 Problem Formulation

The core problem is to rigorously compare and evaluate state-of-the-art hybrid Deep Learning (DL) models for performing **Human ID classification** (identifying 1 of 30 subjects) using the complex, time-series nature of CSI and associated metadata.

Furthermore, to ensure the resulting model is reliable, reproducible, and ready for potential deployment, this research integrates a comprehensive **MLOps pipeline**, allowing for automated tracking, evaluation, and hardware resource monitoring.

1.3 Objective

To design, implement, and evaluate an MLOps pipeline for comparing the performance of multiple specialized hybrid DL models (CSILSTMNet, DenseNet1D, EfficientNet1DLSTM, MobileNetV3_1D_LSTM) in classifying Human IDs based on Wi-Fi CSI data.

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2. Dataset and Preprocessing (Chapter 3 - Section A)

2.1 Wi-Fi Human Activity and ID Dataset

The research utilizes the "A dataset for Wi-Fi-based human activity recognition in LOS and NLOS indoor environments". This dataset serves as a benchmark for both Human Activity Recognition (HAR) and **Human ID**.

- **Scope:** The data involves **30 subjects (S1–S30)** (male and female participants, age ~22.7 ± 2.95 years). Biometric information is noted to subtly affect the signal signatures.
- **Data Content:** The dataset provides both **Channel State Information (CSI)** and **RSSI values**.
- **Target Task:** The classification target is **Human ID**, involving up to 30 unique classes (subjects). The model configuration in the code suggests aiming for 31 classes (`num_classes=31`).

2.2 Data Loading and Sequence Generation

The custom `WifiCSIDataset` class manages the data pipeline:

1. **Feature Extraction:** Raw CSV files are loaded. Data is partitioned into:
    - **CSI Features:** 99 complex CSI values, converted to magnitudes.
    - **Metadata Features:** 12 features, including `timestamp_low`, `bfee_count`, `Nrx`, `Ntx`, RSSI values (`rssi_a`, `rssi_b`, `rssi_c`), `agc`, and permutation details.
2. **Scaling:** To normalize the time series data across the entire dataset, **StandardScaler** is applied independently to both the CSI data and the metadata features.
3. **Windowing:** Data sequences are generated using a **sliding window approach**. The parameters defined in the data loading sequence are a **window size of 128** and a **stride of 64**. This temporal segmentation is crucial for training sequence models like LSTMs.

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3. Deep Learning Methodology and MLOps Pipeline (Chapter 3 - Section B & C)

3.1 Hybrid Deep Learning Architectures

The project performs a comparative study across four distinct DL architectures, all designed to process both the time-series CSI sequence and the auxiliary metadata sequence simultaneously.

Model Name	CSI Processing Core	Metadata Processing Core	Key Features	Source Citation
CSILSTMNet	Standard LSTM	Standard LSTM	Uses the last hidden state from two parallel	

			LSTMs (one for CSI, one for Meta).	
DenseNet1D	Dense Block 1D CNN	Bidirectional LSTM	Uses 1D CNN blocks (DenseBlock1D) followed by global pooling for CSI, combined with a Bi-LSTM for metadata.	
EfficientNet1DLSTM	MBConv1D Blocks	Bidirectional LSTM	Leverages **Mobile Inverted Bottleneck (MBConv)** blocks (stem, mbconv1-3) designed for efficiency, combined with a Bi-LSTM.	
MobileNetV3_1D_LSTM	MobileNetV3 Block 1D	Bidirectional LSTM	Incorporates specialized blocks (including depthwise separable convolutions and Squeeze-and-Excitation (SE) layers) optimized for mobile/efficient computing, combined with a Bi-LSTM.	

3.2 MLOps Implementation with MLflow

The research uses an MLOps pipeline (`CSI_ID_MLOPS_advanced.py`) to manage and track the extensive training process.

1. **Configuration Management:** Parameters such as `epochs=100`, `learning_rate=0.0005`, and model configuration details (`resnet_filters`, `lstm_units`) are managed via a dedicated `ConfigReader` class reading from `csi_id_config.properties`.
2. **Experiment Tracking (MLflow):** MLflow is utilized to manage the experiment runs (`experiment_name = 100_EPOCHS_EXP_06_04102025`) and ensure reproducibility.
   - **Hyperparameter Logging:** All parameters, including model architecture inputs, learning rates (e.g., `0.001, 0.0005`), and epoch counts are logged per run.
   - **Performance Metrics:** Extensive metrics are logged **per epoch**, including:
     - Training and Validation Loss/Accuracy (`val_accuracy`, `val_loss`).
     - Precision, Recall, and F1 Score (weighted average).
     - Area Under the Curve (AUC) for multi-class classification.
3. **Hardware Monitoring:** A critical aspect of MLOps is tracking resource usage. Metrics logged per epoch include **CPU Utilization Percent**, **Memory Used GB**, and **Disk Usage GB**.
4. **Model and Artifact Management:**

- The model with the **best validation accuracy** (`best_val_acc`) is saved as a checkpoint (`.pt` file).
- MLflow logs artifacts including the checkpoint file, plots of Accuracy/Loss curves, and evaluation reports.
- Evaluation artifacts include the **Confusion Matrix (`cm`)** and the detailed **Classification Report (`cr_report`)** computed on the test set using the best saved model.

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4. Expected Results and Discussion (Chapter 4)

4.1 Comparative Model Performance

The evaluation phase will compare the four hybrid models based on their ultimate performance metrics (Validation Accuracy, F1 Score, AUC), identifying the model that offers the most robust and accurate classification for passive human ID.

The results will discuss which components—the streamlined MobileNetV3/EfficientNet architectures or the denser CNN/LSTM structures—prove more effective at extracting individual-specific movement signatures from the CSI data.

4.2 MLOps Insights

The detailed logging of resource consumption (CPU, Memory, Disk) will provide key insights into the operational efficiency of the models, which is crucial for real-world MLOps deployment considerations. For example, comparing the resource intensity of **DenseNet1D** versus the potentially more lightweight **MobileNetV3_1D_LSTM** will inform deployment decisions.

The MLOps framework ensures that the selection of the best model is based not just on accuracy, but on a holistic view of performance across various hyperparameters and computational resources.